

National University of Singapore
School of Computing
CS1101S: Programming Methodology
Semester I, 2026/2027

S9
Mutable Data

Problems:

1. Write the function `d_filter` that takes as arguments a one-argument predicate function `pred` and a linked list `xs`, and returns a linked list that contains only those elements for which `pred` returns `True`. Your function must not create any new pair, and the result linked list must only be made of existing pairs in `xs`. Your function must not modify the head of any of the existing pairs.

```
def d_filter(pred, xs):  
    # ???
```

Example call:

```
L = llist(1, 2, 3, 4, 5, 6, 7, 8, 9, 11)  
d_filter(lambda x: x % 2 == 0, L) # returns [2, [4, [6, [8, None]]]]  
L # What is L now?
```

